

Lecture 5a

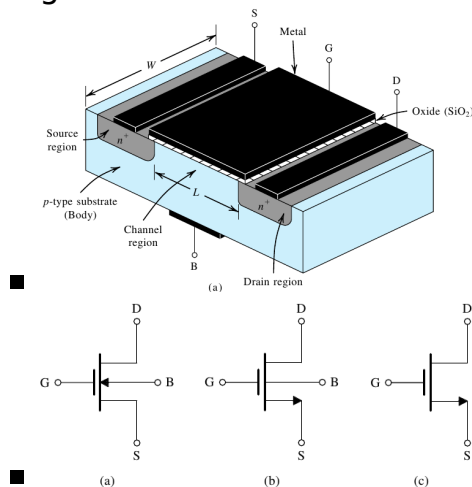
EE-215 Electronic Devices and Circuits

Asst Prof Muhammad Anis Ch

Current-Voltage Characteristics

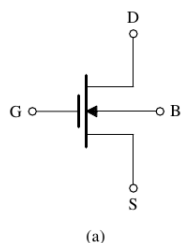
Circuit Symbol

- the circuit symbols for an n-channel enhancement type MOSFET are shown here
 - the two vertical lines represent the gate and the channel
 - the spacing between the two lines indicate that the gate electrode is insulated from the substrate



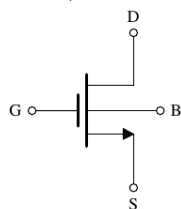
- the arrow indicates the polarity of the substrate (p-type) and the channel (n-type)
 - the arrow points from the p-type substrate to the n-type channel

- thus the arrow indicates that this is an n-channel MOSFET



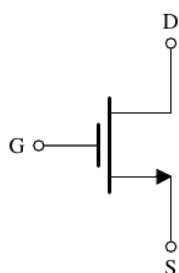
- when it is required to designate one terminal as the source and the other as the drain, this symbol is more convenient
 - here the arrow head is placed at the source terminal
 - and the arrowhead points in the direction of current flow

- recall that for an n-channel MOSFET, electrons flow from the source to the drain
- $\Rightarrow$ current flows from the drain to the source



■ (b)

- when the source is connected to the body of the device,
 - a simplified circuit symbol can be used

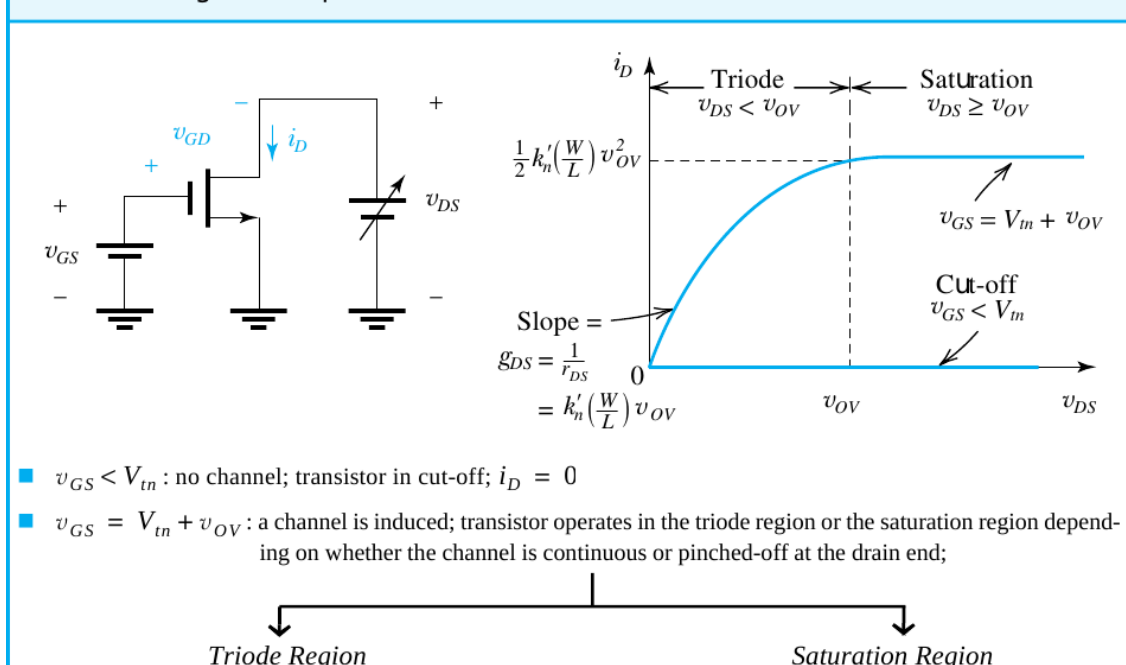


■ (c)

The $i_D - v_{DS}$ Characteristics

- for an NMOS transistor, there are three possible regions of operation
 - the cutoff region
 - the triode region
 - the saturation region

Table 5.1 Regions of Operation of the Enhancement NMOS Transistor



- $v_{GS} < V_{tn}$: no channel; transistor in cut-off; $i_D = 0$
- $v_{GS} = V_{tn} + v_{OV}$: a channel is induced; transistor operates in the triode region or the saturation region depending on whether the channel is continuous or pinched-off at the drain end;

Triode Region

Continuous channel, obtained by:

$$v_{GD} > V_{tn}$$

or equivalently:

$$v_{DS} < v_{OV}$$

Then,

$$i_D = k'_n \left(\frac{W}{L} \right) \left[(v_{GS} - V_{tn}) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

or equivalently,

$$i_D = k'_n \left(\frac{W}{L} \right) \left(v_{OV} - \frac{1}{2} v_{DS} \right) v_{DS}$$

Saturation Region

Pinched-off channel, obtained by:

$$v_{GD} \leq V_{tn}$$

or equivalently:

$$v_{DS} \geq v_{OV}$$

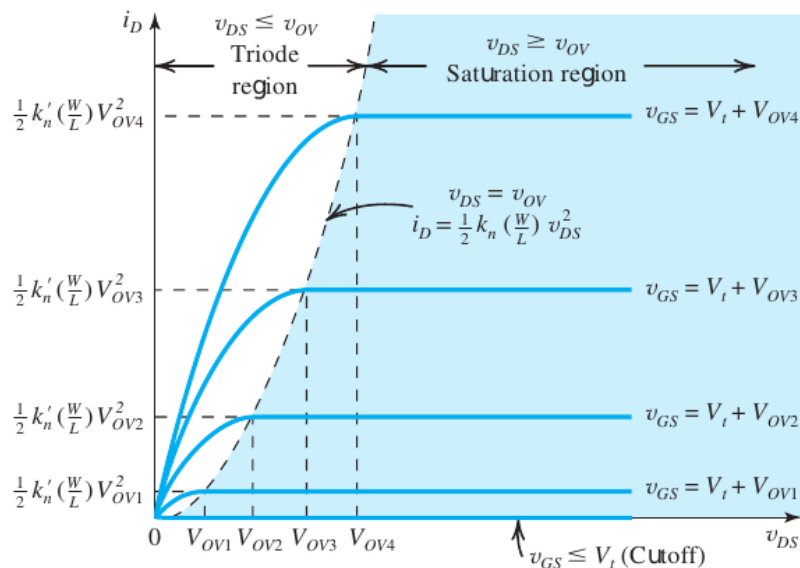
Then

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2$$

or equivalently,

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{OV}^2$$

- $i_D - v_{DS}$ characteristic for the NMOS transistor with different v_{GS} settings is shown in fig



○ **Figure 5.13** The $i_D - v_{DS}$ characteristics for an enhancement-type NMOS transistor.

- Note that the boundary between the triode and the saturation regions, can be given as

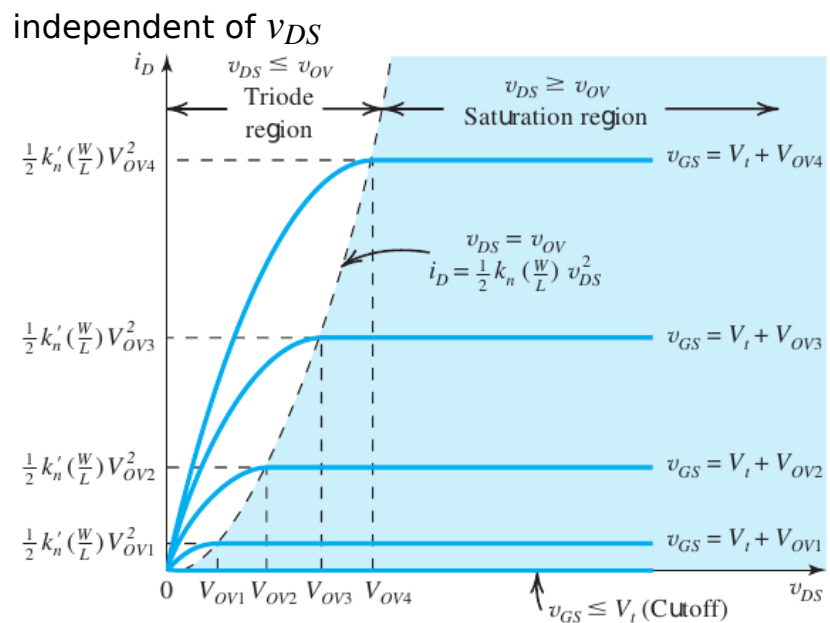
$$○ i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{DS}^2$$

$$■ \because i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) v_{OV}^2 \text{ and}$$

$$■ v_{OV} = v_{DS} \text{ at the start of saturation region}$$

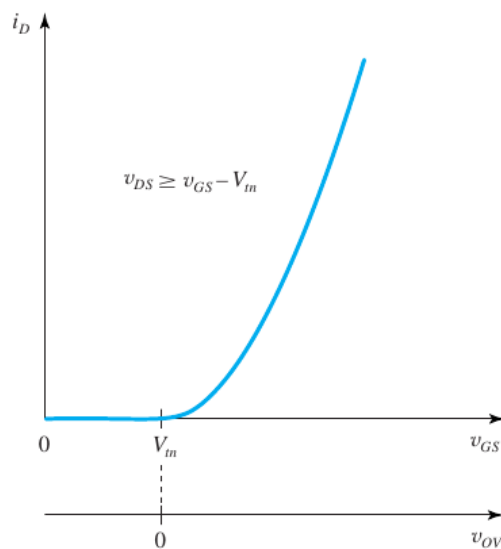
The $i_D - v_{GS}$ Characteristics

- A MOSFET operating in the saturation region, can be used to design an amplifier
 - in saturation, the drain current is a constant (i.e. does not change with v_{DS})
 - and is determined by v_{GS} (or v_{OV}) and is



■ **Figure 5.13** The i_D - v_{DS} characteristics for an enhancement-type NMOS transistor.

- $\Rightarrow$ the MOSFET operates as a constant-current source, whose current is determined by v_{GS} (or v_{OV})
- thus the MOSFET is a voltage controlled current source with the control relationship given as
- $i_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2$ or $i_D = \frac{1}{2} k_n' \left(\frac{W}{L} \right) v_{OV}^2$
- the $i_D - v_{GS}$ characteristics of an NMOS transistor operating in saturation region can be given as



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- thus the MOSFET in the saturation region acts as a voltage-controlled current source
- the MOSFET in the saturation region acts as
 - a voltage-controlled current source and can be modeled as

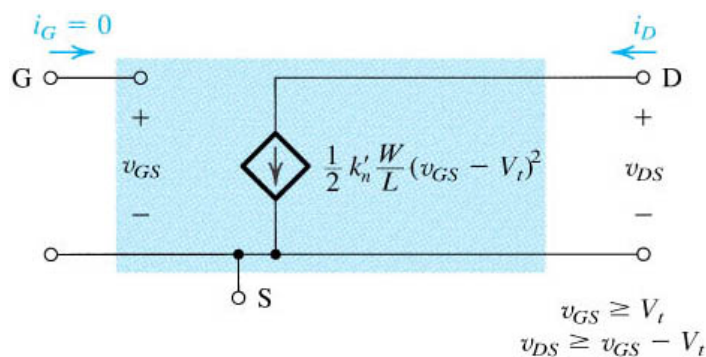


Figure 4.13 Large-signal equivalent-circuit model of an n -channel MOSFET operating in the saturation region.

- this equivalent circuit is known as large signal equivalent circuit model
- Note that the current source is ideal i.e. it has an infinite output resistance
- $= \Rightarrow$ which represents that i_D is independent of v_{DS} in saturation region.
- In practice, this output resistance is finite because of channel-length modulation.

Finite Output Resistance in Saturation

- Uptill now, we have assumed that in saturation, i_D is independent of v_{DS}

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2$$

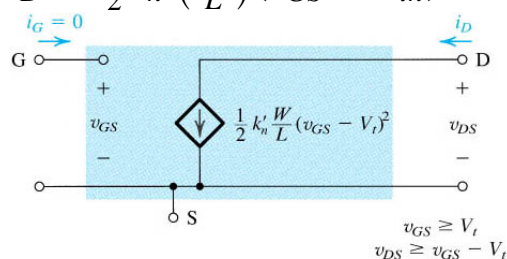


Figure 4.13 Large-signal equivalent-circuit model of an n -channel MOSFET operating in the saturation region.

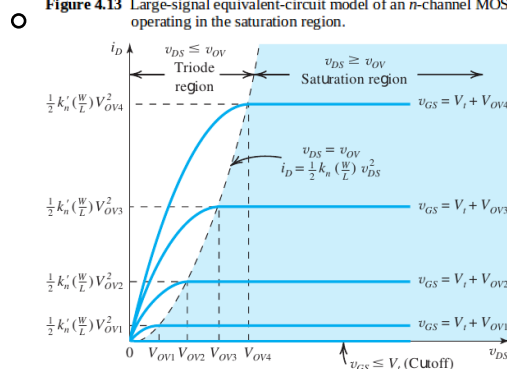


Figure 5.13 The i_D - v_{DS} characteristics for an enhancement-type NMOS transistor.

- i.e. a change Δv_{DS} in the drain to source voltage causes a zero change in i_D
- $= \Rightarrow$ that the incremental resistance looking into the drain of a saturated MOSFET is infinite
- this is not true in practice

- As v_{DS} is increased, the channel pinch-off point is
 - moved slightly away from the drain, towards the source.

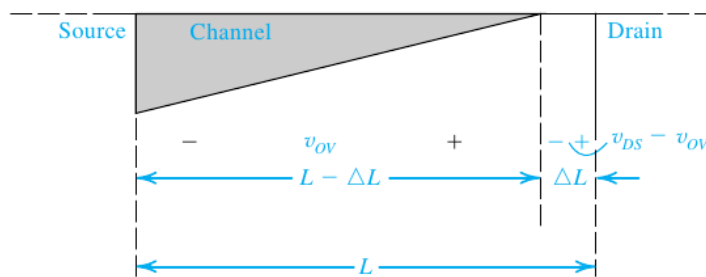


Figure 5.16 Increasing v_{DS} beyond v_{DSsat} causes the channel pinch-off point to move slightly away from the

- drain, thus reducing the effective channel length (by ΔL).
- Note that the voltage across the channel still remains constant at v_{OV}
- Note that the voltage across the channel still remains constant at v_{OV} ,
 - while the additional voltage applied to the drain appears as a voltage across
 - the depletion region between the end of the channel and the drain
- so the channel length is now reduced from L to $L - \Delta L$
- and this phenomenon is called channel-length modulation
- As i_D is inversely proportional to the channel length
 - $i_D = \frac{1}{2}k_n' \left(\frac{W}{L} \right) (v_{GS} - V_{tn})^2$ for $v_{DS} = v_{OV}$
 - $\Rightarrow i_D$ increases with v_{DS} as L decreases with v_{DS} because of channel-length modulation
 - To account for the dependance of i_D on v_{DS} in saturation,
 - we replace L in the above equation by $L - \Delta L$
 - $\Rightarrow i_D = \frac{1}{2}k_n' \left(\frac{W}{L - \Delta L} \right) (v_{GS} - V_{tn})^2$
 - multiply and divide by L
 - $i_D = \frac{1}{2}k_n' \frac{W}{L} \left(\frac{L}{L - \Delta L} \right) (v_{GS} - V_{tn})^2$
 - $i_D = \frac{1}{2}k_n' \frac{W}{L} \left(\frac{L - \Delta L}{L} \right)^{-1} (v_{GS} - V_{tn})^2$
 - $i_D = \frac{1}{2}k_n' \frac{W}{L} \left(1 - \frac{\Delta L}{L} \right)^{-1} (v_{GS} - V_{tn})^2$
 - $i_D = \frac{1}{2}k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 \left(1 - \frac{\Delta L}{L} \right)^{-1}$
- $i_D = \frac{1}{2}k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 \left(1 - \frac{\Delta L}{L} \right)^{-1}$
 - assuming $\Delta L < L \Rightarrow \frac{\Delta L}{L} < 1$
 - As for $|x| < 1$, $(1 \pm x)^n \approx 1 \pm nx$

$$\blacksquare = \Rightarrow \left(1 - \frac{\Delta L}{L}\right)^{-1} \approx 1 - (-1) \frac{\Delta L}{L} = 1 + \frac{\Delta L}{L}$$

$$\circ i_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 \left(1 + \frac{\Delta L}{L}\right)$$

$$\circ \text{ as } \Delta L \text{ increases by increasing } v_{DS} = \Rightarrow \Delta L \propto v_{DS}$$

$$\blacksquare \text{ let the proportionality constant be } \lambda'$$

$$\blacksquare = \Rightarrow \Delta L = \lambda' v_{DS} \text{ and } \frac{\Delta L}{L} = \frac{\lambda' v_{DS}}{L} = \lambda v_{DS}$$

$$\circ = \Rightarrow i_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 (1 + \lambda v_{DS})$$

$$\circ \text{ where } \lambda \text{ is a device parameter having the units of } V^{-1}$$

$$\circ \text{ the value of } \lambda \text{ depends on the process technology and on the channel length } L$$

- because of this channel-length modulation,
 - there is a linear dependence of i_D on v_{DS} in the saturation region

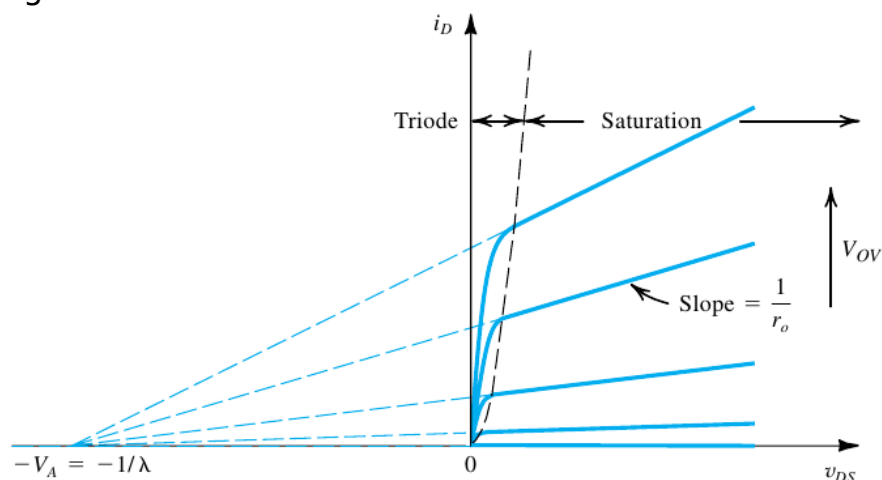


Figure 5.17 Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L .

- if these straight lines are extrapolated,
 - they all intersect the v_{DS} axis at the same point $v_{DS} = -V_A$,
 - where V_A is a positive voltage called Early voltage, $V_A = \frac{1}{\lambda}$
 - V_A is a device parameter. its value is dependent on the process and the channel length.
 - i.e. $V_A = V_A' L$ where V_A' is entirely process technology dependent
- thus because of the channel length modulation
 - value of i_D depends on v_{DS} , in the saturation region
 - so for a given v_{GS} , a change Δv_{DS}
 - $= \Rightarrow$ corresponding change Δi_D in the drain current i_D

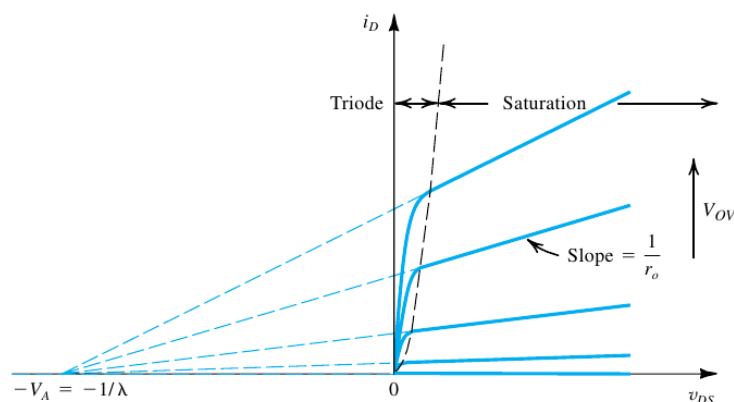


Figure 5.17 Effect of v_{DS} on i_D in the saturation region. The MOSFET parameter V_A depends on the process technology and, for a given process, is proportional to the channel length L .

- $\Rightarrow$ the output resistance of the current source representing i_D in saturation

- is no longer infinite. but is a finite value represented by r_o

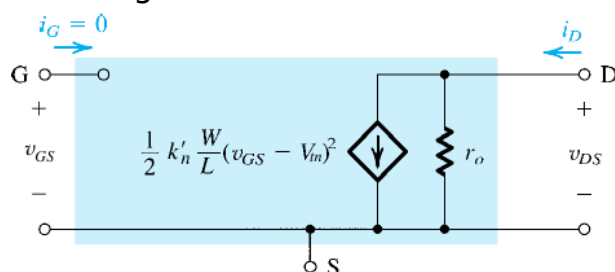


Figure 5.18 Large-signal equivalent circuit model of the n -channel MOSFET in saturation, incorporating the output resistance r_o . The output resistance models the linear dependence of i_D on v_{DS} and is given by Eq. (5.23).

- where $r_o = \left[\frac{\partial i_D}{\partial v_{DS}} \right]_{v_{GS}=\text{constant}}^{-1}$
- $r_o = \left[\frac{\partial i_D}{\partial v_{DS}} \right]_{v_{GS} \text{ constant}}^{-1}$
 - $r_o = \left[\frac{\partial}{\partial v_{DS}} \left\{ \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 (1 + \lambda v_{DS}) \right\} \right]_{v_{GS} \text{ constant}}^{-1}$
 - $r_o = \left[\frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 \frac{\partial}{\partial v_{DS}} (1 + \lambda v_{DS}) \right]^{-1}$
 - $r_o = \left[\frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 (0 + \lambda(1)) \right]^{-1}$
 - $r_o = \left[\frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 \lambda \right]^{-1}$
 - $r_o = \frac{1}{\lambda} \left[\frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2 \right]^{-1} = \frac{1}{\lambda} [I_D]^{-1}$
 - $r_o = \frac{1}{\lambda I_D} = \frac{V_A}{I_D}$
 - where I_D is the drain current without channel-length modulation effect i.e. $I_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_{tn})^2$
- $r_o = \frac{1}{\lambda I_D} = \frac{V_A}{I_D}$
 - $\Rightarrow r_o \propto \frac{1}{I_D}$

- thus the large-signal equivalent circuit model incorporating the channel-length modulation i.e. r_o can be given as

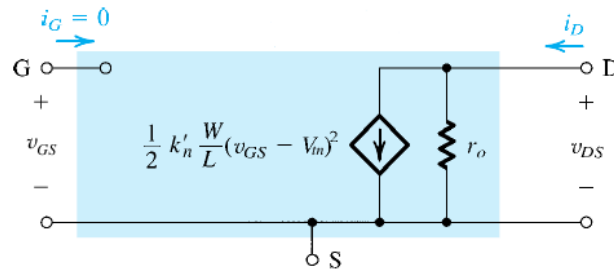
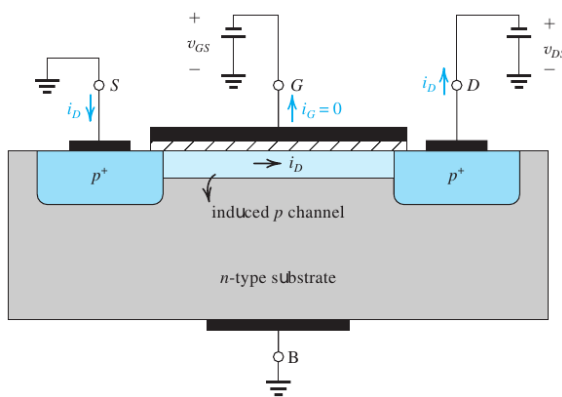


Figure 5.18 Large-signal equivalent circuit model of the n -channel MOSFET in saturation, incorporating the output resistance r_o . The output resistance models the linear dependence of i_D on v_{DS} and is given by Eq. (5.23).

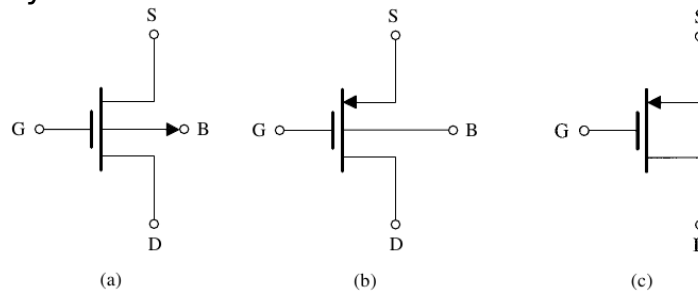
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Characteristics of the p-Channel MOSFET

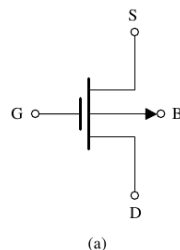


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- the p-channel enhancement type MOSFET, has the circuit symbol of



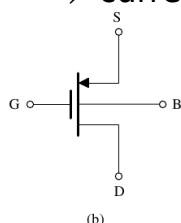
- the arrow indicates that the channel is p-type (tail of arrow)
- while the substrate is n-type (head of arrow)



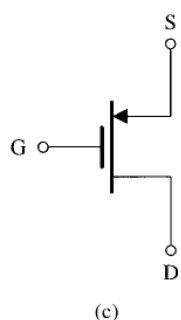
- this symbol is utilized when it is required to designate the source terminal
- the arrow head is at the source and its direction indicates the

direction of current flow

- Note that in a p-channel MOSFET holes flow from source to the drain
- $\Rightarrow$ current flows from source to drain



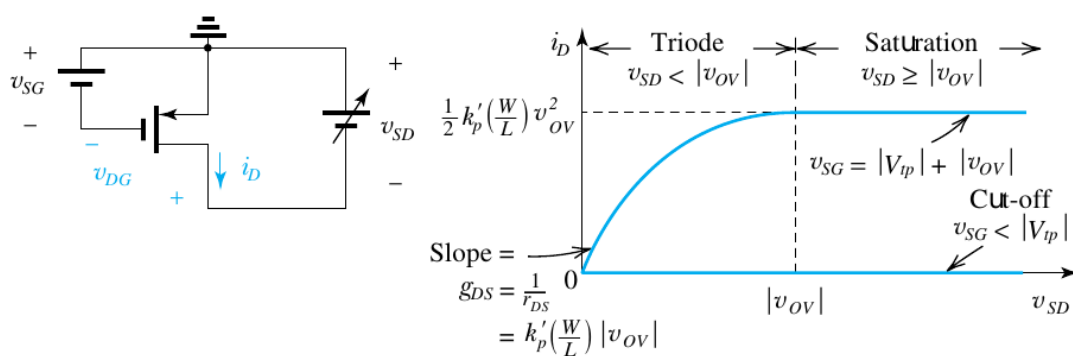
- for the case where the source is connected to the substrate,
 - this simplified symbol is used



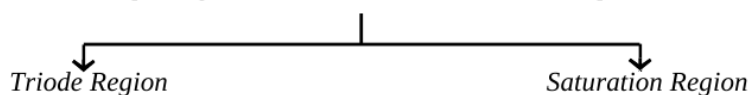
Current-Voltage Characteristics

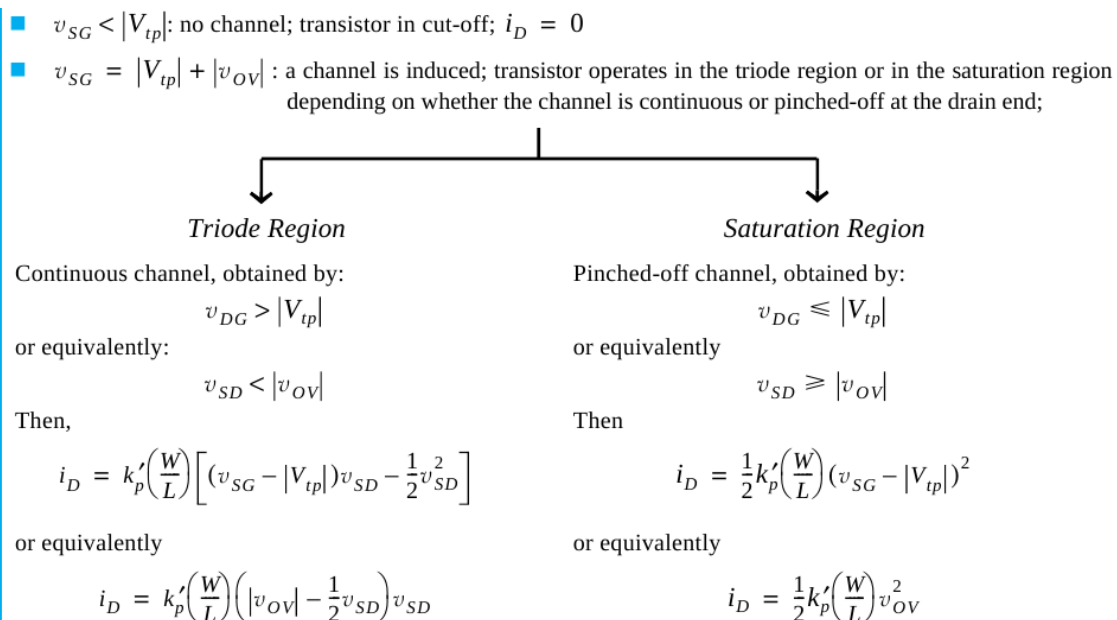
- the regions of operation of the PMOS transistor are

Table 5.2 Regions of Operation of the Enhancement PMOS Transistor



- $v_{SG} < |V_{tp}|$: no channel; transistor in cut-off; $i_D = 0$
- $v_{SG} = |V_{tp}| + |v_{OV}|$: a channel is induced; transistor operates in the triode region or in the saturation region depending on whether the channel is continuous or pinched-off at the drain end;





- just like the NMOS transistor, the PMOS device also suffers from Channel-length modulation effect.
 - thus the saturation region expression for i_D can be given as
 - $i_D = \frac{1}{2} k_p' \left(\frac{W}{L} \right) (v_{SG} - |V_{tp}|)^2 (1 + |\lambda| v_{SD})$
 - and $\lambda = \frac{1}{V_A}$
 - where λ and V_A for the PMOS transistor are negative quantities
 - Note that for a given CMOS fabrication process, in general
 - $\lambda_n \searrow = |\lambda_p|$ and $V_{An} \searrow = |V_{Ap}|$